

PART 1 - GENERAL

1.1 Summary

26 32 13 Part 1.1.1.A This section specifies the requirements for standby diesel engine generator sets for the Meridian Data Centre Campus, Phase 1. The generator plant shall provide emergency power to the full campus load including IT systems, cooling, and building services during utility power interruptions.

1.2 References

26 32 13 Part 1.2.1.A The following standards shall apply: ISO 8528 (Reciprocating Internal Combustion Engine Driven Alternating Current Generating Sets), IS 10000 (Methods of Tests for IC Engines), IS 13364 (Gas/Dual Fuel Engine Generator Sets), BS 5514 (Reciprocating IC Engines Performance), NFPA 110 (Standard for Emergency and Standby Power Systems).

1.3 Submittal Requirements

26 32 13 Part 1.3.1.A Submittals shall include engine performance data sheets, alternator test certificates, fuel consumption curves at 25%, 50%, 75%, and 100% load, ambient derating curves, noise emission data, exhaust emission data, and dimensional drawings with weights.

1.4 Quality Assurance

26 32 13 Part 1.4.1.A Generator set manufacturer shall have ISO 9001 certification and a minimum of 10 years experience in manufacturing standby generator sets rated above 1500 kVA for mission-critical installations.

1.5 Warranty

26 32 13 Part 1.5.1.A Generator set warranty shall be a minimum of 24 months from the date of commissioning or 2000 operating hours, whichever occurs first. Warranty shall cover engine, alternator, control system, and all factory-supplied accessories.

PART 2 - PRODUCTS

2.1 Acceptable Manufacturers

26 32 13 Part 2.1.1.A Acceptable generator set manufacturers: Deccan Diesel Co., or approved equivalent meeting all performance requirements of this section.

2.2 Performance and Design Criteria

26 32 13 Part 2.2.1.A Generator standby power rating shall be a minimum of 2500 kVA. The standby rating shall be verified and guaranteed at the project design basis ambient temperature of 45°C. Ratings quoted at lower reference ambient temperatures shall be derated per the manufacturer's published derating curve.

26 32 13 Part 2.2.1.B After applying the manufacturer's published ambient temperature derating factors for operation at 45°C, the available standby output shall not fall below 2500 kVA. The vendor shall submit the derating curve and demonstrate compliance at design ambient.

26 32 13 Part 2.2.1.C Generator power factor rating shall be 0.8 lagging. Generator shall be capable of delivering rated kW output continuously at 0.8 power factor.

26 32 13 Part 2.2.1.D Generator output voltage shall be 11 kV, 3-phase, 50 Hz. Voltage regulation shall be within +/- 0.5% under steady-state conditions.

26 32 13 Part 2.2.1.E Generator operating speed shall be 1500 RPM synchronous for 50 Hz output. Speed governing shall meet ISO 8528-5 performance class G3 requirements.

26 32 13 Part 2.2.2 Generator shall accept 100% block load in a single step and recover to steady-state voltage and frequency within the limits specified by ISO 8528-5 performance class G3. Transient voltage dip shall not exceed 15% and recovery to within 3% shall occur within 10 seconds.

26 32 13 Part 2.2.3.A Fuel consumption rate at 100% standby load shall not exceed 520 litres per hour of high speed diesel (HSD) conforming to IS 1460. The vendor shall submit a fuel consumption certificate from the engine manufacturer.

26 32 13 Part 2.2.4.A Fuel consumption rate at 75% load shall be submitted in the vendor performance data table.

26 32 13 Part 2.2.4.B Fuel consumption rate at 25% load shall be submitted in the vendor performance data table.

26 32 13 Part 2.2.4.C The vendor shall submit a complete performance data table listing engine output power, alternator output, fuel consumption, exhaust temperature, and cooling water temperature at 25%, 50%, 75%, and 100% load points.

26 32 13 Part 2.2.4.D Fuel consumption rate at 50% load shall be submitted in the vendor performance data table. Omission of any load point data from the performance table shall be treated as a non-compliant submittal.

2.3 Component Description

26 32 13 Part 2.3.1.A Engine shall be a multi-cylinder, turbocharged, charge air-cooled, 4-stroke diesel engine designed for standby duty. Engine shall conform to CPCB Stage IV emission norms or equivalent.

26 32 13 Part 2.3.2.A Engine speed governor shall be electronic type, meeting ISO 8528-5 performance class G3 requirements. Governor shall provide isochronous frequency control suitable for paralleling applications.

26 32 13 Part 2.3.3.A Generator shall be equipped with dual starter motors in redundant configuration, each with independent 24 VDC starter batteries and battery chargers. Automatic crank sequence shall attempt start on the primary starter, and transfer to the secondary starter after three failed attempts.

26 32 13 Part 2.3.4.A Alternator shall be brushless, self-excited, with a permanent magnet pilot exciter. Insulation class shall be Class H (180°C) with Class F (155°C) temperature rise at rated load.

26 32 13 Part 2.3.5.A Generator controller shall be the current model EMCP 4.4 digital electronic control panel, or approved equivalent, providing engine monitoring, protection, and paralleling control functions. Obsolete controller versions shall not be accepted.

26 32 13 Part 2.3.6.A Radiator cooling system shall be engine-driven with belt-driven fans. Cooling system shall be designed for continuous operation at 45°C ambient temperature.

26 32 13 Part 2.3.7.A Exhaust system shall include a hospital-grade residential silencer providing a minimum of 25 dBA attenuation. Exhaust piping shall be insulated and lagged to limit surface temperature to below 60°C.

26 32 13 Part 2.3.8.A Bulk fuel storage system shall consist of 4 underground double-walled steel tanks of 100,000 litres each, providing 48 hours of fuel autonomy for the complete generator plant at full standby load.

26 32 13 Part 2.3.8.B Fuel transfer pumping system shall include duty and standby transfer pumps with automatic changeover. The fuel transfer system shall maintain the local day tank level between the high and low level setpoints at all times.

26 32 13 Part 2.3.8.C Each generator shall be provided with a local day tank sized to provide a minimum of 2.0 hours of continuous operation at full standby load. Based on the specified fuel consumption rate of 520 litres per hour, the minimum day tank capacity shall be 1040 litres.

2.4 Physical Construction and Acoustic Treatment

26 32 13 Part 2.4.1.A Generator set base frame shall be heavy-duty fabricated steel with integral anti-vibration mounts. Base frame shall include a fuel-tight drip tray.

26 32 13 Part 2.4.2.A Generator set enclosure (if outdoor installation) shall be weatherproof to IP44 with acoustic lining to meet the specified noise limits.

26 32 13 Part 2.4.3.A Generator set noise level shall not exceed 105 dBA at 1 metre distance without acoustic enclosure. Manufacturer shall state the open-set and enclosed noise levels separately.

26 32 13 Part 2.4.3.B With the acoustic enclosure and silencer installed, the generator set sound pressure level shall not exceed 85 dBA measured in a free field at 1 metre distance from the enclosure surface at full rated standby load.

PART 3 - EXECUTION

3.1 Examination and Preparation

26 32 13 Part 3.1.1.A Verify that generator room foundations, exhaust penetrations, fuel piping, and electrical cable routes are complete and ready prior to generator delivery.

3.2 Installation

26 32 13 Part 3.2.1.A Generator sets shall be rigged and placed on prepared foundations using certified lifting equipment. Anti-vibration mounts shall be adjusted to achieve level installation within 2 mm across the base frame.

26 32 13 Part 3.2.1.B Flexible fuel connections, exhaust bellows, and cable gland plates shall be installed per manufacturer details. All fuel connections shall be leak tested at 1.5 times working pressure.

26 32 13 Part 3.2.1.C Battery banks and chargers shall be installed and commissioned prior to engine first start. Electrolyte levels shall be verified in flooded-type batteries.

26 32 13 Part 3.2.1.D Cooling water system shall be filled, vented, and leak tested. Coolant mixture shall be per engine manufacturer's recommendation for the local climate conditions.

26 32 13 Part 3.2.1.E All generator room cable trays, busbar trunking, and power cable terminations shall be completed and torque-marked prior to energisation of the generator output circuit breaker.

26 32 13 Part 3.2.1.F Adequate ventilation shall be provided in the generator room to prevent heat buildup and ensure proper engine combustion air supply during full load operation. Room ventilation provisions shall be verified complete prior to engine commissioning.

3.3 Startup and Field Testing

26 32 13 Part 3.3.1.A Prior to initial start, perform a complete pre-start checklist including oil level, coolant level, belt tension, battery voltage, and control system self-test.

26 32 13 Part 3.3.2.A Perform initial no-load start test. Engine shall reach rated speed and voltage within 10 seconds of the start command.

26 32 13 Part 3.3.3 Perform initial start sequence and voltage balance checks for engine 3. Verify three-phase voltage balance is within 1% of nominal at no load and at 50% load.

26 32 13 Part 3.3.4 Perform initial start sequence and voltage balance checks for engine 4. Verify three-phase voltage balance is within 1% of nominal at no load and at 50% load.

26 32 13 Part 3.3.4.A Perform 100% block load acceptance test in a single step per ISO 8528-5 performance class G3. Record transient voltage dip, frequency dip, and recovery times. Results shall meet the limits specified in Part 2.2.2 of this section.

26 32 13 Part 3.3.5 Perform initial start sequence and voltage balance checks for engine 5. Verify three-phase voltage balance is within 1% of nominal at no load and at 50% load.